Import Tariffs, Illegal Immigration and Tariff Threats

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Abstract

Using a two-country specific factors model of trade and illegal immigration we show that factor market terms of trade losses and increased illegal immigration weaken the case for an import tariff for the developed host nation of immigrants. We also show that free trade is not Pareto efficient in the presence of illegal immigration and immigration related costs. Finally, we consider tariff threats by the host nation of immigrants designed to induce emigration-deterring border enforcement by the source nation. Compared to a no-threat equilibrium, we show that a tariff threat strategic equilibrium can lower illegal immigration and raise the host nation's welfare.

JEL Codes: F1, F2

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1. Introduction

International trade is well understood to generate gains for participating nations through greater access to global markets. The other side of this coin is international movement of productive factors such as labor and capital, which also confer gains by improving the global allocation of resources. International labor mobility, however, carries social and political implications for nations that go beyond pure allocative efficiency considerations. Rapid immigration into a rich host nation from a poorer source nation, with differences in language, social norms, and institutions, can impose assimilation related costs on both newcomers and incumbent residents. Additional pressures may arise from congestion of publicly provided goods when population density increases and tax revenues do not scale up accordingly.[1] Keeping such costs in mind and weighing them against benefits conferred by a productive immigrant work force, nations calibrate their immigration policy, often through the use of quantitative entry limits or quotas.

Immigration quotas are typically country- and skill-specific, leaving large real wage differentials between developed host nations (e.g., USA) and developing source nations (e.g., Mexico). These wage gaps create strong incentives for illegal immigration from the source to the host nations despite existing migration barriers. A central barrier is border protection measures by the developed nation, but such measures can be costly for the host nation due to both economic and non-economic reasons. Indeed, the ineffectiveness of such measures in stemming recent illegal immigration flows to the US has been a matter of serious policy concern. The post-Covid surge in illegal immigration has, in part, prompted the US administration not only to strengthen border operations but also to employ tariff threats to induce greater cooperation from Mexico in controlling the bilateral border. There are two central questions that arise in this context. First, what is the effect of an import tariff on wages and welfare levels in

[1] For a discussion of publicly provided goods in the context of illegal immigration see Bandyopadhyay and Pinto (2017) and Pena (2014), among others.

the tariff imposing nation (say US) and on the illegal immigration flow between the two nations (say US and Mexico)? Second, can such a tariff be strategically used by the US to raise its own welfare? We discuss in the next two paragraphs how sections 2 and 3 of the paper address these questions. A word of caution is appropriate for the reader at this point. The objective of the exercise is not to derive "optimal" tariff threats, but simply to assess possible welfare implications of a particular type of tariff threat under several simplifying assumptions, including non-retaliation by the trade partner and a particular assumed sequence of moves.

Section 2 presents a two-good (good 1 and good 2) competitive specific factors model of trade between two countries A and B, coupled with equilibrium illegal migration of labor from the poorer nation B (say Mexico) to the richer nation A (say the US).[2] Both goods are produced by both nations, however in a trading equilibrium the excess of A's domestic demand over domestic production of good 1 is imported from nation B. In exchange, A exports good 2 to nation B. An import tariff on good 1 imposed by A raises the tariff-inclusive domestic price of good 1 in A, reducing import demand.[3] The fall in import demand by A leads to an excess supply of good 1 in the international market, leading to a fall in the international market clearing price of good 1. Since good 1 is A's import good, the fall in the international price of good 1 confers a goods/output market terms of trade gain to nation A. On the factor market side, an import tariff by A raises the domestic price in A and pushes up the domestic wage in A (through an increase in the value of marginal product of labor) attracting more illegal immigration. Both the rise in illegal immigration and a rise in the wage paid to migrants (factor market terms of

[2] We abstract from a multi-good, multi-country analysis for the sake of simplicity and analytical tractability. The production structure of the two goods in both countries follows standard specific factors modeling assumptions. Labor is the common factor of production for the two goods, while each good employs a distinct factor specific to the production of that good. As an example, consider good 1 to be a manufactured good, and good 2 to be an agricultural good, where good 1 is produced by labor and capital, where capital is specific to the production of good 1, and where good 2 is produced by labor and land, where land is specific to the production of good 2.

[3] Notice that there are two prices of good 1 to consider here. One is the international price or border price p (before tariff is applied) and the other is the tariff inclusive price faced by the importing nation's consumers (and producers), which is $p+t$, where t is a unit import tariff on good 1 imposed by nation A.

trade loss) reduce *A*'s welfare. These latter welfare costs oppose the goods market terms of trade benefit of a tariff for *A* and weaken the case for the use of such a tariff for welfare improvement. However, all of this assumes that *A* cannot strategically influence *B* by using tariffs as a threat rather than actually imposing it. This is where the analysis of section 3 comes in as we discuss next.

Section 3 considers the case where nation *A* uses tariff threats to incentivize *B* to increase border enforcement to reduce illegal emigration from *B* to *A*. To model a threat rather than the outright choice of the tariff itself, we consider a situation where policies are chosen in three stages. Nation *A* makes a tariff threat in stage 1 that unless *B* chooses to boost its border enforcement in stage 2 to a desirable level for *A*, *A* will impose tariffs in stage 3. We show that if *A* is committed to carrying out its tariff threat, then the threat can incentivize *B* to choose *A*'s desired border enforcement level in stage 2. What is critical here is that *B* must believe that *A* will actually carry out its tariff threat, and section 3 discusses when such a threat is credible. We show that an equilibrium of such a sequential game can confer higher gains to *A* because it does not in reality impose any distortionary tariffs (because *B* does what *A* wants), yet *A* gets reduced illegal immigration because *B* raises enforcement against emigration. Nation *B* will likely be worse off (compared to no-threat situation) because of increased border enforcement cost but will not have any incentive to depart from this equilibrium unless it can use alternate tools like tariff retaliation.

Ethier (1986) provides a classic analysis of the interaction between illegal immigration flows and border interdiction and how they jointly affect the welfare of a developed host nation. Subsequent papers, nicely summarized by Hanson (2006, see pages 915-916), build on Ethier's framework by incorporating capital mobility, trade, foreign aid, and other factors into the study of illegal immigration. Within this tradition, Bandyopadhyay (2006) examines the illegal immigration and welfare effects of a preferential reduction of tariff by one "small" nation on its

bordering "small" nation which is the source nation of immigrants (while keeping tariffs unchanged on the rest of the world). We draw on some elements of Bandyopadhyay's (2006) modeling approach, which uses dual functions in the tradition of Dixit and Norman (1980). However, the context of the current paper differs substantially because we focus on a two-country environment with endogenous determination of output market prices and a strategic sequential setting involving tariff threats. The literature on simultaneous trade and factor mobility suggests that trade taxes can be used to benefit from output market terms of trade coupled with taxes on wages or on capital imports to benefit from the factor market terms of trade. A classic example of such an analysis is Srinivasan (1983). Our paper differs from this literature by considering the public costs of immigration which interact with the welfare effects of trade policy, and also through our consideration of tariff threats in a strategic setting which lead to some novel findings regarding both illegal immigration control and welfare effects.

It is important to note that free trade and free labor mobility may not be jointly optimal for the two nations because assimilation cost of immigrants is a central friction which, if sufficiently large, renders the free-market solution inefficient.

Section 2 presents the basic model, which allows us to characterize the simultaneous choice of border enforcement and trade policies by the host nation while the source nation of immigrants maintains free trade. Section 3 analyzes a three-stage sequential game where the developed nation employs a tariff threat to induce the source country to strengthen border control measures and reduce emigration. Section 4 concludes.

2. The Model

We present below a two-good specific factors framework which can be used to analyze the interlinkages between import tariffs and illegal immigration flows between two nations *A* and *B*. Nation *A* is assumed to be a high wage nation attracting illegal immigration from nation *B* which

has a lower wage rate. Nation *A* produces and consumes both goods 1 and 2, while nation *B* consumes the numeraire good 2 only, but produces both goods. Therefore, good 1 is a pure export good for nation *B* (i.e., good 1 is nation *A*'s import good, on which *A* potentially applies a unit import tariff at rate *t*) which it trades to import good 2.[4] We also assume that nation *B* is committed to free trade.[5]

As in standard competitive models of trade, we assume that competitive profit maximizing firms produce each of the goods using constant returns technology. Good 1 is produced by labor and some factor specific to the production of good 1, and similarly good 2 is also produced by labor and a factor specific to the production of good 2. Labor is assumed to be perfectly mobile within each nation between the sectors producing goods 1 and 2, respectively. The specific factor for each sector is assumed to be internationally immobile. The natives of each nation are represented by a representative consumer for that nation. They are assumed to be atomistic and they seek to maximize their utility given their respective budget constraints. Under these assumptions, the production/supply side and the demand side of each nation can be represented by a nation's *Revenue function* and *Expenditure function*, respectively. We describe these functions and their important properties in the next subsection. Finally, we assume in this section that nation *A*'s government uses border enforcement to maximize its natives' welfare. Under these assumptions, this section analyzes the effect of an import tariff by *A* on the welfare of each of the two nations and on the joint welfare of the two nations.

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[4] The assumed production and consumption structure are helpful in streamlining the analysis of terms-of-trade and welfare in a model of labor mobility. To be clear, when we refer to nation *j*'s consumption we are referring to the consumption of nation *j*'s natives. For example, under these assumptions, immigrants, who are from nation *B*, do not consume good 1. This allows us to easily define the demand for good 1 without having to worry about how the demand for good 1 may be affected by the division of *B*'s citizens between emigrants and non-emigrants.

[5] This assumption is in the tradition of a textbook optimal tariff scenario for a large country (say nation *A*) which may benefit from a terms-of-trade improving import tariff in the absence of retaliation by the trade partner (nation *B* in the current context). In practice, a trade partner may not retaliate depending on the relative bargaining positions of the nations involved, non-economic factors like common membership in defense alliances etc. which are beyond the scope of the current paper.

2.1 Revenue and Expenditure Functions

This subsection introduces some functions that have been used in dual general equilibrium models of trade along the lines of Dixit and Norman (1980). In an economy with given factor endowments, perfectly competitive profit maximizing firms, and constant return to scale in production, the supply side of the economy can be summarized by the *Revenue function*. This function is the sum of revenues obtained by all the producing firms. Alternately, under constant returns to scale production functions, the revenue function represents the sum of all factor earnings in a nation. Using Dixit and Norman (1980), we know that such a revenue function for nation *j* takes the form $R^j\left(p^{1j}, v^j\right)$, $j = A, B$, where p^{1j} is the tariff inclusive relative price of good 1 in nation *j* and v^j is nation *j*'s endowment vector.[6] It can be shown that the revenue function is convex in prices and concave in factor endowments. Since we will only consider changes in factor endowments through labor migration, we use v^j and v^{j*} to denote nation *j*'s pre- and post-migration labor endowments, respectively. Standard envelope properties of the revenue functions yield the post-migration supply function of good 1 in nation *j* as $R_p^j\left(p^{1j}, v^{j*}\right)$.[7] In addition to providing us with the supply function of good 1, the revenue function also defines the wage rate in nation *j*. In particular, the wage rate in nation *j* is simply the partial of the revenue function with respect to the post-migration labor endowment of the nation, such that the wage rate in nation *j* is $w^j = R_v^j\left(p^{1j}, v^{j*}\right)$.

[6] Notice that the analysis focuses only on the market for good 1 because by Walras law the market for good 2 must be in equilibrium at the market clearing price of good 1.

[7] We use the following notation: for any function $f(x, y)$, f_x denotes first partial of the function with respect to *x*, f_{xx} the second partial with respect to *x*, and f_{xy} denotes the partial of f_x with respect to *y*.

Notice that nation A imposes the unit import tariff t on good 1 while nation B follows free trade, such that the respective tariff inclusive prices for good 1 in the two nations are $p+t$ and p, respectively. Turning to the factor endowments, notice that an immigration flow I from B to A increases A's labor supply to $v^{A*} = v^A + I$, while reducing B's labor supply to $v^{B*} = v^B - I$. At these labor supply levels in the two nations, their respective revenue functions are $R^A\left(p+t, v^A + I\right)$ and $R^B\left(p, v^B - I\right)$. Accordingly, the supply of good 1 from nations A and B are $R_p^A\left(p+t, v^A + I\right)$ and $R_p^B\left(p, v^B - I\right)$, respectively. Wage rates in the two nations are $w^A = R_v^A\left(p+t, v^A + I\right)$ and $w^B = R_v^B\left(p, v^B - I\right)$, respectively.

Finally, we turn to the demand side of the economy. Assuming identical preferences among natives of nation A, the nation's expenditure function is $E^A\left(p+t, u^A\right)$, where u^A is the nation's utility/welfare level. Therefore, using standard properties of expenditure functions, the Hicksian demand for good 1 by nation A is given by $E_p^A\left(p+t, u^A\right)$. Given that nation B is assumed not to consume good 1, A's demand for good 1 summarizes the relevant demand side of the model.

2.2 Illegal Immigration and the Equilibrium Migration Condition

We discuss incentives for illegal immigration, the equilibrium migration condition and the level of equilibrium illegal immigration in this subsection. At the outset of section 2 we had assumed that nation A is the high-wage nation attracting labor inflow from nation B, such that we have $w^A = R_v^A\left(p+t, v^A + I\right) > w^B = R_v^B\left(p, v^B - I\right)$, at $t \geq 0$ and $I = 0$. Now, consider an immigration assimilation cost $c(I)$ incurred by the host nation A from a level of immigration I, and assume that this cost is increasing and convex in I, such that $c'(I) > 0$, $c''(I) > 0$. For analytical

simplicity, we assume that this assimilation cost is large enough for nation A such that the nation does not want any immigration.[8] Therefore, any immigration that occurs is illegal immigration. We return later in the next subsection to a discussion of our exclusive focus on illegal immigration, the substitute/complement relationship between immigrant labor and native factors of production, and labor market frictions that immigrants may face.

Along the lines of Ethier (1986) we assume that the host nation A uses border enforcement to contain illegal immigration. For a level of border enforcement a, the probability of a successful border crossing by a potential immigrant is $0 < \psi(a) < 1$, $\psi'(a) < 0$, and $\psi''(a) > 0$, such that border enforcement reduces the successful border crossing probability but at a diminishing rate (i.e., $\psi''(a) > 0 \Leftrightarrow \frac{d|\psi'(a)|}{da} < 0$). If the potential immigrant is caught at the border, then she returns to nation B and earns wage w^B after paying a penalty of k to defray the costs associated with the return process.[9] A risk-neutral potential migrant equates the expected wage from attempting migration to the certain wage w^B available in the source nation, such that the equilibrium migration condition is:

$$\psi w^A + (1-\psi)(w^B - k) = w^B \Rightarrow w^B = w^A - \frac{k(1-\psi)}{\psi}. \tag{1}$$

Recalling that $w^A = R_v^A(p+t, v^A + I)$ and $w^B = R_v^B(p, v^B - I)$, Eq. (1) may be written as:

$$R_v^B(p, v^B - I) = R_v^A(p+t, v^A + I) - \frac{k[1-\psi(a)]}{\psi(a)}, \tag{2}$$

[8] To be clear, consider a hypothetical situation where illegal immigration is not possible, such that the only immigration policy issue for nation A is whether to allow some legal immigration, say $\bar{I}$. In this context, the marginal immigrant assimilation cost for A is $c'(\bar{I})$. If $c'(\bar{I})$ at $\bar{I} = 0$ is sufficiently large, then $(du^A / d\bar{I})_{|\bar{I}=0} < 0$, driven by the high marginal assimilation cost. Thus, for a sufficiently high marginal immigrant assimilation cost, nation A will not want any immigration at all.

[9] Since this penalty simply covers the costs of the return process, it is a deadweight loss and does not feature in either nation's revenue collections.

such that the migration equilibrium condition implicitly defines the illegal immigration level as $I = I(p,t,a)$, such that the level of illegal immigration is determined by the border price of good 1 (i.e., p), the import tariff level t and the border enforcement level a. Notice that the price level p is endogenous and is determined by the equilibrium in the market for good 1, whose analysis we present in section 2.4 after a discussion in the next sub-section of some of our migration-related modeling choices. We also note here that the effects of the policy variables a and t on the illegal immigration level are presented in section 2.5.

2.3 A discussion of the possible coexistence of legal and illegal immigration, substitutability between native and immigrant labor, labor market frictions facing immigrants and a two-country environment

The main issue at hand in this analysis is the use of trade policy as an illegal immigration deterrence tool beyond the usual choice of border enforcement by the host nation of immigrants. Ruling out the possibility of any legal immigration lends greater focus and clarity to the analysis of this central issue. One may wonder how allowing for coexistence of legal and illegal immigration affects the analysis. To consider this question suppose that assimilation costs are more moderate than we have assumed such that some immigration $\overline{I}$ is desirable for the host nation. Notice, however, that as long as the host nation's wage exceeds the source nation wage at the immigration level $\overline{I}$, there will be incentives for more immigration. This additional immigration (say I) beyond the legal limit of $\overline{I}$ will be illegal immigration, with the total immigration level at a migration equilibrium being $\tilde{I} = \overline{I} + I$. In such an environment, constrained optimal levels of $\overline{I}$ and border enforcement policies need to be jointly chosen keeping in mind the fact that immigration assimilation costs depend on total immigration $\tilde{I} = \overline{I} + I$. Several other factors, including trade policy, are at play in this richer model, and it is

hard to make hard predictions without a full analytical treatment. The basic insights from our stripped-down model, however, regarding how tariffs affect illegal immigration and how tariff threats affect welfare are unlikely to be lost in such a more complex analysis.

Our analysis assumes perfect substitutability between native labor and immigrant labor. One may wonder how imperfect substitutability may change the analysis. To answer this question, assume that immigrant labor is an imperfect substitute for native unskilled labor. In this case, a higher native wage will lead employers to substitute some immigrants for natives. This will drive native wages lower and immigrant wages higher until all output and factor markets clear and the equilibrium migration condition (described later) is satisfied. Thus, similar to our model where native wages and immigrant wages are equalized under equilibrium migration, the wage gap between native unskilled labor and immigrant labor is narrowed at equilibrium when the two types of labor inputs are imperfectly substitutable. Overall, the presence of other imperfectly substitutable or complementary factors in the endowment vector of the revenue function should not alter our basic results.

Let us now consider possible labor market frictions faced by illegal immigrants. A nice example of such an analysis is provided by Bond and Chen (1987) which considers natives and illegal immigrants to be perfect substitutes in the production function, but immigrant employees face a probability of detection from "internal enforcement" by immigration authorities leading to a fine on the employers. Thus, a competitive firm hires an illegal immigrant only when the wage of the illegal immigrant plus the expected fine on the employer equal the wage paid to natives. This "friction" of being an "illegal immigrant" leads in equilibrium to a lower wage for the illegal immigrant compared to a native. Analysis of such a model will involve exploring joint optimality of border and internal enforcement and the interactions with all the other policy variables. Again, the central insights about inadequacy of border enforcement measures to stem

illegal immigration and the possibility of reliance on tariff threats explored in this paper are unlikely to be qualitatively altered.

Another assumption invoked to streamline the analysis is the restriction to a two-country world. Given that the U.S. problem of illegal immigration is mostly concentrated along its southwestern border with Mexico, our focus on a single border seems reasonable. However, there may be multiple source countries, and one can plausibly think of migrants from other origins transiting through Mexico before entering the United States. Additionally, a source country may have access to alternative export destinations, allowing it to divert trade away from the host country in response to a tariff. Such an outside option would weaken the host country's ability to use tariffs as a policy instrument to influence a source nation of immigrants. While additional considerations would arise in these cases, the basic insights of the two-nation, one-border model used here should nevertheless remain qualitatively valid. We turn next to the equilibrium migration relationship.

2.4 The market clearing price of good 1 and the effects of changes in the illegal immigration level and in nation A's import tariff on the market clearing price

This subsection describes the equilibrium in the market for good 1, and how the market clearing border price of good 1 (i.e., p) is affected by the import tariff rate t and the illegal immigration level I. The Hicksian demand for good 1 in A is $E_p^A\left(p+t,u^A\right)$, where we assume quasi-linear preferences, such that $E_{pu}^A=0$. Nation B is assumed to only consume good 2, implying that the total demand for good 1 is $E_p^A\left(p+t,u^A\right)$. Turning to the supply side, the total supply of good 1 from the two nations is the sum of $R_p^A\left(\cdot\right)$ and $R_p^B\left(\cdot\right)$. Thus, the market clearing equation for good 1 is:

$$E_p^A\left(p+t,u^A\right)=R_p^A\left(p+t,v^A+I\right)+R_p^B\left(p,v^B-I\right). \tag{3}$$

Eq. (3) implicitly defines the price of good 1 as:

$$p=p\left(t,I\right), \tag{4}$$

Where $p_t=\dfrac{E_{pp}^A-R_{pp}^A}{D}<0$, because $E_{pp}^A<0$ and $R_{pp}^A>0$ because of concavity and convexity of the expenditure and revenue functions in output prices, and because $D=R_{pp}^A+R_{pp}^B-E_{pp}^A$, the slope of the excess supply function of good 1, is strictly positive. Consider why p_t, the effect of the import tariff on the price of good 1, is negative. An increase in the tariff rate *t* raises the domestic price of good 1 in *A* (= $p+t$) and reduces demand. On the other hand, the rise in the domestic price in *A* raises supply of good 1 by producers in nation *A* as they move up along their supply curve given by $R_p^A\left(p+t,v^A+I\right)$. Thus, an increase in the tariff rate reduces demand and raises supply at the initial equilibrium border price (say $p=\tilde{p}$). The resulting excess supply of good 1 at $p=\tilde{p}$ is resolved by an appropriate decline in the market clearing price of good 1 to a new level (say p^*, where $p^*<\tilde{p}$).

Turning to the effect of immigration on the market clearing border price, we use the implicit function rule in Eq. (3) to get $p_I=\dfrac{R_{pv}^B-R_{pv}^A}{D}>0$ if and only if $R_{pv}^B>R_{pv}^A$. To understand these inequalities first note that $R_{pv}^j=\dfrac{\partial R_p^j}{\partial v}$ is the effect of an increase in the labor endowment in nation *j* on the production of good 1 by nation *j*. In a specific factors model an increase in the labor endowment necessarily raises the output of both goods, such that $R_{pv}^j>0$. Now, as illegal immigration *I* rises, *A*'s labor endowment rises, and hence its output of good 1 rises to the tune of R_{pv}^A. On the other hand, the rise in emigration from *B* reduces *B*'s labor endowment and reduces

the supply of good 1 in *B* to the tune of R_{pv}^{B}. When $R_{pv}^{B} > R_{pv}^{A}$, the fall in production of good 1 in *B* exceeds the rise in production of good 1 in *A*, such that the overall supply of good 1 in the international market falls, driving up the equilibrium price of good 1 (i.e., $p_I > 0$). Thus, a rise in illegal immigration raises the price of good 1 if and only if $R_{pv}^{B} > R_{pv}^{A}$.

2.5 The effect of the import tariff and border enforcement on illegal immigration

The relationship between the equilibrium illegal immigration level and border enforcement *a* on the one hand and the import tariff *t* on the other can be obtained by substituting the function $p(t,I)$ defined in Eq. (4) for the variable *p* in Eq. (2). After the substitution, Eq. (2) implicitly defines

$$I = I(a,t), \tag{5}$$

where $I_a = k\psi' / \left(\psi^2 D^I\right) < 0$, $I_t = \left[R_{vp}^{A}\left(p_t + 1\right) - R_{vp}^{B} p_t\right] / D^I > 0$, and

$D^I = \dfrac{\left(R_{vp}^{B} - R_{vp}^{A}\right)^2}{D} - R_{vv}^{A} - R_{vv}^{B} > 0$, because $\psi'(a) < 0$, $R_{vp}^{j} > 0$, $p_t < 0$, and $p_t + 1 > 0$.[10] Thus, using Eq. (5) we know that a rise in border enforcement will reduce illegal immigration, while a rise in the tariff rate will raise illegal immigration. We explain in the next two paragraphs the intuition behind these findings.

A rise in the border enforcement level raises the probability of border detection and hence the expected cost of migration. At this higher enforcement level, the expected return from migration, which is $\psi w^A + (1-\psi)\left(w^B - k\right)$, will fall below the certainty wage w^B violating the equilibrium migration condition. Equilibrium migration will require fewer immigrants trying to

[10] Notice that $R_{vv}^{j} = \dfrac{\partial R_v^j}{\partial v} = \dfrac{\partial w^j}{\partial v} < 0$, because a larger endowment of labor must reduce a nation's wage rate in a specific factors model. Also, we know from Eq. (4) that $p_t < 0$, and can show using Eq. (4) that $p_t + 1 > 0$.

migrate and hence a lower equilibrium level of illegal immigration. We turn next to considering the effect of the tariff on illegal immigration.

Recall from Eq. (4) that an increase in the import tariff imposed by nation *A* reduces the equilibrium border price *p* (i.e., $p_t < 0$). Since nation *B* pursues free trade the border price is the same as the domestic price in nation *B*, such that a tariff induced fall in *p* reduces nation *B*'s wage rate w^B.[11] The tariff, however raises nation *A*'s domestic price (i.e., $p+t$) and in turn raises the wage rate in nation *A*.[12] The consequent rise in the wage differential $\left(w^A - w^B\right)$ drives up illegal immigration, as can be seen from the expression for I_t following Eq. (5). We turn next to the analysis of the effect of the tariff on two nations' utility levels.

2.6 The effect of the tariff on the two nations' utility levels under optimal border enforcement by nation A

This subsection discusses the effects of the policy variables of nation *A*, the import tariff rate *t* and the border enforcement level *a*, respectively, on each nation's utility and on the joint utility of the two nations. We also outline the first order condition (FOC) for optimal choice of border enforcement by nation *A*. Following the tradition of the trade and immigration literature, we will assume that the host nation is concerned only about its own citizens' utility and does not include the income earned by immigrants in its welfare calculations. Keeping this in mind, the budget constraint of the nation as a whole (equivalently its trade balance condition) equates the total expenditure of the nation to its total production revenues and import tax revenues less border

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[11] Notice that $w^B = R_v^B\left(p, v^B - I\right)$, such that given *I*, $\frac{\partial w^B}{\partial t} = R_{vp}^B p_t < 0$, because $R_{vp}^B > 0$ and $p_t < 0$.

[12] The domestic price $p+t$ in nation *A* must rise because $\frac{\partial\left(p+t\right)}{\partial t} = p_t + 1 > 0$. In turn, this means that a higher tariff must raise *A*'s wage rate, $w^A = R_v^A\left(p+t, v^A + I\right)$, because given *I*, $\frac{\partial w^A}{\partial t} = R_{vp}^A\left(p_t + 1\right) > 0$, where $R_{vp}^A > 0$.

protection costs $\theta(a)$, the assimilation cost incurred by the hosting nation $c(I)$, and immigrant wage payments Iw^A. Thus, A's expenditure-earnings relationship is:

$$E^A\left[p(t,I)+t,u^A\right]=R^A\left[p(t,I)+t,v^A+I\right]+tM^A-Iw^A-c(I)-\theta(a), \tag{6}$$

where $I=I(a,t)$, $w^A=R_v^A\left\{p\left[t,I(a,t)\right]+t,v^A+I(a,t)\right\}$, and $M^A=E_p^A-R_p^A$, is the difference between A's domestic demand and domestic supply of good 1, equaling the nation's import of good 1. Eq. (6) implicitly defines nation A's utility level as $u^A=u^A(a,t)$. Noting that E_u^A is the inverse of the marginal utility of income and is unity for quasi-linear utility, the FOC for the optimal border enforcement level is:

$$u_a^A=-M^A p_I I_a-c'I_a+t\frac{\partial M^A}{\partial a}-I\frac{\partial w^A}{\partial a}-\theta'(a)=0, \tag{7}$$

where the first term on the right-hand side (RHS) of Eq. (7) is terms of trade effect through the influence of reduced immigration on the price of good 1. The second term is the welfare gain from savings in assimilation costs of immigrants. The third term is the amplification (or reduction) of the tariff's distortionary cost through an import volume effect of enforcement. The fourth term is the factor market term of trade effect through changes in w^A and the last term is the marginal cost of enforcement.

Recall that Eq. (6) defines the utility of nation A, $u^A=u^A(a,t)$. Using the latter utility function we will now explore the effect of an increase in the import tariff t on nation A's utility, assuming that border enforcement is optimally set [i.e., assuming that u_a^A in Eq. (7) is set to zero]. Differentiating $u^A(a,t)$ and using $u_a^A=0$, we get

$$\frac{du^A}{dt}=u_t^A=-M^A p_t-I(1+p_t)\frac{\partial w^A}{\partial p}-\frac{\theta'}{|I_a|}I_t+t\left(E_{pp}^A-R_{pp}^A\right)(1+p_t). \tag{8}$$

The first term on the RHS of Eq. (8) is the standard terms of trade gain effect for A from a tariff induced decline in the import price p (recall $p_t < 0$). The second term, inclusive of the minus sign, reflects the factor market terms of trade loss because immigrants are now paid a tariff induced higher wage. The third term reflects the utility loss due to a larger inflow of immigrants due to the tariff (under optimal border enforcement). The last term is the deadweight losses from standard production and consumption distortions caused by the tariff. Evaluating Eq. (8) at free trade we get:

$$\left(\frac{du^A}{dt}\right)_{|t=0} = u_t^A\left(t=0\right) < 0 \Leftrightarrow -M^A p_t < \left(1+p_t\right) I \frac{\partial w^A}{\partial p} + \frac{\theta'}{\left|I_a\right|} I_t, \tag{9}$$

such that a small tariff reduces A's welfare (starting from free trade) if the terms of trade gain in the output market is outweighed by the sum of the factor market terms of trade loss and the utility loss from increased immigration. We assume from this point that A's disutility from immigration is sufficiently strong to ensure that $u_t^A\left(t=0\right) < 0$, such that free trade is unilaterally optimal for nation A.

We will now consider the effect of A's tariff on joint welfare of the two nations and on the welfare of nation B and establish that free trade cannot be a jointly efficient outcome. For that purpose, we first define nation B's utility net of migration costs. Nation B's utility is given by its consumption of good 2, which must equal the sum of B's revenues and wage receipts from A net of fines paid by the migrants caught at the border. Suppose that I^* is the number of individuals attempting immigration. Given that $1-\psi$ fraction of potential migrants fails to migrate and incurs a per head cost of k, the total migration cost is $k\left(1-\psi\right)I^*$. Now, the number

of successful migrants is $I = \psi I^* \Rightarrow I^* = I/\psi$. Thus, total migration cost incurred by nation *B* is $\frac{k(1-\psi)I}{\psi}$. Therefore, nation *B*'s utility net of migration costs is:[13]

$$u^B = c_2^B = R^B\left(p, v^B - I\right) + w^A I - \frac{k(1-\psi)}{\psi} I = R^B\left(p, v^B - I\right) + w^B I, \tag{10}$$

where the last equality obtains by noticing that $w^B = w^A - \frac{k(1-\psi)}{\psi}$ from Eq. (1). Using Eqs. (6) and (10) the sum of tariff induced utility changes in the two nations is:

$$\left[\frac{d\left(u^A + u^B\right)}{dt}\right]_{|t=0} = \left(\frac{Ik\psi'}{\psi^2} - \theta'\right)\frac{da}{dt} - c'\frac{dI}{dt} < 0, \tag{11}$$

if $\frac{da}{dt} > 0$ and $\frac{dI}{dt} > 0$. Both the latter inequalities are likely to hold because a rise in tariff will raise nation *A*'s wage and encourage immigration, which will in turn incentivize greater border enforcement "*a*".[14] Eq. (11) shows that joint welfare must fall when a tariff is used because the tariff induced immigration raises *A*'s assimilation costs and border enforcement costs; while also raising resource costs for *B* in terms of penalties incurred by potential emigrants caught at the border. Thus, Eq. (11) shows that free trade cannot be a jointly efficient outcome. Indeed, a small import subsidy that reduces immigration increases joint welfare. However, given the likely infeasibility of an import subsidy which will need to be funded by taxes, we focus on an alternative instrument in the next section to identify conditions under which these deadweight

[13] For simplicity of exposition, we abstract from possible reduction of congestion cost for nation *B* coming from emigration.

[14] If the marginal cost of enforcement curve $\theta'(a)$ is very steep at the enforcement optimum, then $\frac{da}{dt} \cong 0$. In this case, $\frac{dI}{dt} = I_a \frac{da}{dt} + I_t \cong I_t$, such that Eq. (11) reduces to $\left[\frac{d\left(u^A + u^B\right)}{dt}\right]_{|t=0} = -c' I_t < 0$.

losses may be reduced. Finally, note that because $u_a^A = 0$, $\frac{du^A}{dt} = u_t^A$. Thus, using Eq. (11), we can also get the effect of the tariff on nation *B*'s utility as:

$$\left(\frac{du^B}{dt}\right)_{|t=0} = -u_t^A + \left(\frac{Ik\psi'}{\psi^2} - \theta'\right)\frac{da}{dt} - c'\frac{dI}{dt}, \tag{12}$$

Where u_t^A at free trade is given by Eq. (8) evaluated at $t = 0$.

3. Tariff threat and border enforcement by both nations

This section considers the implications of a tariff threat by nation *A* to incentivize border enforcement by nation *B* and thereby reduce illegal immigration. For that purpose, as a benchmark case, section 3.1 considers *B*'s incentives to choose border enforcement in the absence of any tariffs or tariff threats by nation *A*. Subsequently, section 3.2 discusses the issue of credibility of tariff threats by *A*. Finally, section 3.3 provides an analysis of a credible and equilibrium tariff threat which can raise *A*'s welfare by reducing illegal immigration while preserving free trade.

Consider that in addition to border enforcement at level *a* by nation *A*, nation *B* uses border enforcement at level *b* to deter emigration. The probability of successful border crossing is now $\psi = \psi(a,b)$, $\psi_a < 0$, $\psi_b < 0$, $\psi_{aa} > 0$, and $\psi_{bb} > 0$,[15] which requires that both nations' border enforcement levels reduce the probability of successful border crossing but do so with diminishing returns.

3.1 Border enforcement choice by nation B under free trade

[15] The sign of the cross partial ψ_{ab} affects potential substitutability/complementarity between the two nations' enforcement efforts. However, as will become clear later in this section, the assumed model structure allows us to abstract from these cross effects.

We consider in this subsection nation *B*'s unilateral incentives for border enforcement under no tariffs or tariff threats by nation *A*. The relationship between illegal immigration on the one hand, and the policy variables *a*, *b* and *t* on the other, can be obtained by using $\psi = \psi(a,b)$ in Eq. (2), and then using Eq. (5) to get:

$$I = I(a,b,t), \tag{13}$$

where $I_a = k\psi_a / (\psi^2 D^I) < 0$, $I_b = k\psi_b / (\psi^2 D^I) < 0$, and $I_t = \left[R_{vp}^A (p_t + 1) - R_{vp}^B p_t\right] / D^I > 0$. As one would expect, the aforementioned inequalities leave the qualitative effects of *A*'s policy variables *a* and *t* on illegal immigration unaltered, while showing that *B*'s border enforcement effort reduces illegal immigration.

Using $I = I(a,b,t)$ in Eq. (6) we can show that:

$$u_b^A = \frac{|I_b| \theta'(a)}{|I_a|} > 0, \tag{14}$$

such that an increase in *B*'s border enforcement raises *A*'s welfare under free trade ($t = 0$).

Let $\phi(b)$ denote *B*'s border enforcement cost, where $\phi'(b) > 0$, and $\phi''(b) > 0$. Eq. (10) is now amended to:

$$u^B = R^B \left\{ p\left[t, I(a,b,t)\right], v^B - I(a,b,t) \right\} + w^B I(a,b,t) - \phi(b) = u^B(a,b,t). \tag{15}$$

Optimal border enforcement by *B* is given by:

$$u_b^B = R_p^B p_I I_b + I \frac{\partial w^B}{\partial b} - \phi'(b) \leq 0. \tag{16}$$

For consistency with section 2, we assume that *B*'s marginal enforcement cost $\phi'(b)$ is sufficiently large such that $u_b^B(b = 0) < 0$, which means that *B*'s optimal choice of *b* is zero.

3.2 Credibility of tariff threat incentivizing border enforcement by source nation of immigrants

This subsection explains how a tariff threat may or may not be credible in a sequential strategic equilibrium. Notice that nation *B* has no interest in using border enforcement to deter emigration; however, Eq. (14) shows that nation *A* stands to gain from positive enforcement effort by *B*. In this context, *A* can gain from designing a tariff scheme that induces *B* to strengthen border enforcement. We discuss next two scenarios where *A* can potentially raise its welfare by using tariff threats to induce stricter border enforcement by *B*. In both scenarios, the game proceeds in three stages. In stage 1, country *A* announces a tariff schedule $t = \gamma(b)$, with $\gamma'(b) < 0$, such that nation *A* reduces the import tariff (from some high tariff level it also announces in stage 1) to reward *B* for choosing a higher border enforcement level. In stage 2, nation *B* chooses *b* taking into consideration *A*'s announced tariff schedule. Finally, in stage 3, *A* either abides by the promised tariff schedule or chooses to deviate.[16] This sequential structure captures the strategic commitment inherent in tariff threats. While real-world policy adjustments may involve multiple rounds of interaction, the sequential formulation isolates the key question: under what conditions can the host nation credibly commit to a contingent tariff policy that alters the source nation's enforcement incentives? A simultaneous-move framework cannot account for this commitment problem. The subgame perfect equilibrium concept ensures that nation *A*'s threat is credible only if it can commit to follow through; without such commitment, nation *B* anticipates deviation and the threat fails.

Case 1: *No cost for A to deviate from the announced tariff schedule.*

[16] In the present analysis, we consider a specific case of a tariff schedule $t = \gamma(b)$, where the function $\gamma(b)$ is differentiable in *b*. In principle, however, this mapping need not even be continuous in *b* (for instance, it could involve threshold tariff threats where *A* promises to wipe out all of *B*'s gains from trade through sufficiently high tariffs if *B* does not choose *A*'s desired level of *b*). The tariff schedule should be interpreted as a stylized representation of the negotiation set. Moreover, our model assumes perfect observability of enforcement effort *b*. In practice, while outcomes are stochastic, policy inputs such as border patrol budgets, staffing levels, and legislative mandates are verifiable.

Working backward, notice that in stage 3, there is no incentive for *A* to impose a positive tariff because *b* has already been chosen in stage 2, and *A*'s unilateral optimal tariff is zero. When *B* moves in stage 2, it recognizes that the optimal choice for *A* in stage 3 is free trade. Consequently, *B* chooses zero border enforcement, given that $u_b^B(b=0)<0$. Anticipating this sequence of outcomes, *A* has no incentive to announce a tariff threat in stage 1. In other words, a credible tariff threat equilibrium does not exist in this case.

Case 2: *Reputation cost for not abiding by the announced tariff schedule*

In this case, we assume that nation *A* values its international reputation. While *A* can deviate from a commitment, doing so jeopardizes how other nations view *A's* credibility in future negotiations.[17] A simple way of modeling such loss of credibility is to assume that *A* incurs a reputation cost, *C*, in stage 3 if it fails to abide by the tariff schedule announced in stage 1.[18] If *C* is sufficiently large then the net payoff for *A* in stage 3 from deviating (and choosing $t=0$) is lower than the payoff from abiding by its announced tariff schedule: $\tilde{u}^A = u^A(t=0) - C < u^A\left[t=\gamma(b)\right]$. The latter inequality ensures that for any feasible *b* (that does not push *B* to autarky), *A* will always abide by the tariff schedule announced in stage 1. In stage 2, nation *B* recognizes that it is in *A*'s interest to stick to the tariff schedule and appropriately chooses *b* to maximize its own welfare. Anticipating this sequence of choices, *A* optimally designs its tariff schedule in stage 1.

3.3 Tariff threat schedule under reputation cost: An example

[17] These costs may also be justified in the context of political economy. Suppose a government is elected on a political platform proposing that trade policy be conditional on border enforcement. Once elected, this government would face political backlash (e.g., failure to be reelected, losing the "rents" from holding office) if it deviates from the announced policy.

[18] While we model reputation costs *C* in reduced form for tractability, this assumption proxies for the incentive compatibility constraints that would arise in an infinite-horizon repeated game.

This subsection provides an example of a tariff schedule that is consistent with Case 2 described above. Using Eq. (15), the stage 2 payoff function of nation *B* is:

$$u^B = u^B\left[a, b, \gamma(b)\right]. \tag{17}$$

Nation *B*'s first order condition for choice of border enforcement is:

$$u_b^B + u_t^B \gamma'(b) = 0, \tag{18}$$

where assuming that *B* loses from *A*'s tariff (i.e., $u_t^B < 0$), *A* chooses a γ' large enough to rule out a corner solution for *B*.[19] One must consider, however, whether *B* will want to play this trade game at all or revert to autarky. The trade participation constraint for *B* is that it achieves at least the autarky utility level, say u^{B0}. Given that $u_b^A > 0$ and $u_t^A < 0$, nation *A* will want the highest possible level of *b* without making *B* quit trading, since a higher *b* not only reduces immigration but also reduces the distortionary tariff, i.e., $\frac{dt}{db} = \gamma'(b) < 0$. Nation *B*'s trade participation constraint is $u^B\left[a, b, t = \gamma(b)\right] \geq u^{B0}$. We assume, for expositional simplicity, that the marginal cost curve for *A*'s enforcement, $\theta'(a)$, is very steep at the initial free trade equilibrium, such that *a* is frozen at the initial optimal level $a = a^*$.[20] Suppressing a^* from *B*'s utility function, the participation constraint reduces to $u^B\left[b, t = \gamma(b)\right] = u^{B0}$, because *A* will push *B*'s utility to the autarky level in its pursuit of greater enforcement by *B*. The most desirable *b* for nation *A* is given by b^0, where $u^B\left[b = b^0, t = \gamma\left(b^0\right) = 0\right] = u^{B0}$, because *A* will want free trade while also ensuring the highest possible *b*. Using Eq. (18) we see that *A* can obtain $\left(b = b^0, t = 0\right)$ by

[19] Notice from Eq. (15) that $u_t^B = R_p^B\left(p_t + p_I I_t\right) + Iw_t^B$. We assume that the output market terms of trade effect of the tariff, $p_t < 0$, is sufficiently strong such that $u_t^B < 0$. If $u_t^B \geq 0$, nation *A* cannot use tariff threats to incentivize *B* to choose a positive enforcement level.

[20] Allowing *a* to vary considerably complicates the analysis by bringing in an additional strategic effect to consider where *a* can potentially be altered to influence *b*.

choosing $\gamma'\left(b^0\right)$, such that $\gamma'\left(b^0\right) = -\dfrac{u_b^B\left(b^0, t=0\right)}{u_t^B\left(b^0, t=0\right)}$. For a specific functional form, consider $\gamma(b) = s\left(b^0 - b\right)$, where s is a positive constant. Now, consider Figure 1, where $u^B(b,t) = \overline{u}^B$ denote indifference curves for nation B. Note that both b and t reduce B's utility, such that indifference curves closer to the origin represent higher utility. The slope of B's indifference curve at any point $(b > 0, t > 0)$ in the figure is given by $-\dfrac{u_b^B(b,t)}{u_t^B(b,t)} < 0$, because the respective marginal utilities are strictly negative. Assuming sufficiently strong increases in marginal disutility from tariffs and enforcement, B's indifference curves are concave to the origin. The autarky indifference curve is $u^B(b,t) = u^{B0}$, where the horizontal intercept is $b = b^0$. By choosing $s = \dfrac{u_b^B\left(b = b^0, t = 0\right)}{u_t^B\left(b = b^0, t = 0\right)}$, we have the tariff line $t = s\left(b^0 - b\right)$, tangent at $\left(b = b^0, t = 0\right)$ to the autarky indifference curve. Given this tariff line, $u^B(b,t) = u^{B0}$ is the best indifference level that B can get to, so that the choice $b = b^0$ and $t = 0$ is an equilibrium.

Notice that the threat of a tariff by A is enough to get a free trade equilibrium with A's desired level of b, given feasibility. The tariff threat function embodies tariff reductions for higher enforcement by B. To gain these tariff concessions B raises its enforcement, which benefits A. By a suitable choice of the tariff schedule A is able to make B choose A's desired enforcement level (on B's side of the border) without reverting to autarky. Such an equilibrium marks a welfare improvement for A compared to section 2 because A's enforcement and tariff levels remain unchanged at $a = a^*$ and $t = 0$, respectively, while B chooses a positive enforcement conferring positive spillover benefits to A. Finally, recall from section 2 that the free trade equilibrium is inefficient because of "too high" a level of immigration. Therefore, it is possible that joint welfare is higher under the threat of the tariff, where the utility cost for B of

positive enforcement may be more than offset by utility gains for *A* from the immigration reduction.

4. Conclusion

We present a simple model of the interaction between trade and border interdiction policy to control illegal immigration. The first analytical section (section 2) considered the effect of an import tariff by the immigrant hosting nation on illegal immigration and welfare levels of the two nations under optimal border enforcement by the host nation. The effect of the tariff on host nation welfare was shown to depend on the interplay of the output and factor market terms of trade, the assimilation costs of immigrants and on the distortions to allocative efficiency caused by the tariff. We also showed that free trade is not necessarily efficient in terms of maximization of joint welfare of the nations.

Section 3 extends the basic model to a strategic context where nation *A* uses a tariff threat to incentivize nation *B* to choose a higher border enforcement (compared to the zero level for nation *B* in section 2) to help reduce illegal immigration. We show that tariff threats cannot be an equilibrium strategy in the absence of a commitment mechanism, such as a reputation cost that forces the developed/host nation of immigrants to eventually carry out its threat. In the presence of such costs, however, we show that a tariff threat can raise both the host nation's welfare and joint welfare while maintaining free trade.[21] We believe this finding is novel to the rapidly evolving literature on illegal immigration.

We acknowledge that this model is stylized, with several assumptions made for the sake of analytical tractability and expositional clarity. These assumptions allow us to isolate the

[21]Notice that if joint welfare is higher under the tariff threat (which is by no means guaranteed), then transfers between the two nations can raise both nations' welfare levels compared to the initial no-threat equilibrium.

central driving forces, specifically output and factor market terms of trade changes, and strategic considerations, and to derive closed-form solutions for optimal trade and enforcement policies. However, this tractability comes at the cost of ignoring other contributing and potentially confounding factors associated with using trade policy to gain leverage in immigration policy. For example, we abstract from the possibility of tariff retaliation by the developing/source nation of immigrants, which could trigger a downward spiral in trade and worsen welfare for both countries. We also abstract from the effect of tariffs on the real wage differential between the host and source nations, which would complicate the migration equilibrium condition.

On balance, we believe the clarity of the analysis and the insights justify the choice of this framework. Extensions that can generalize and build upon the current model are on our agenda for future research.

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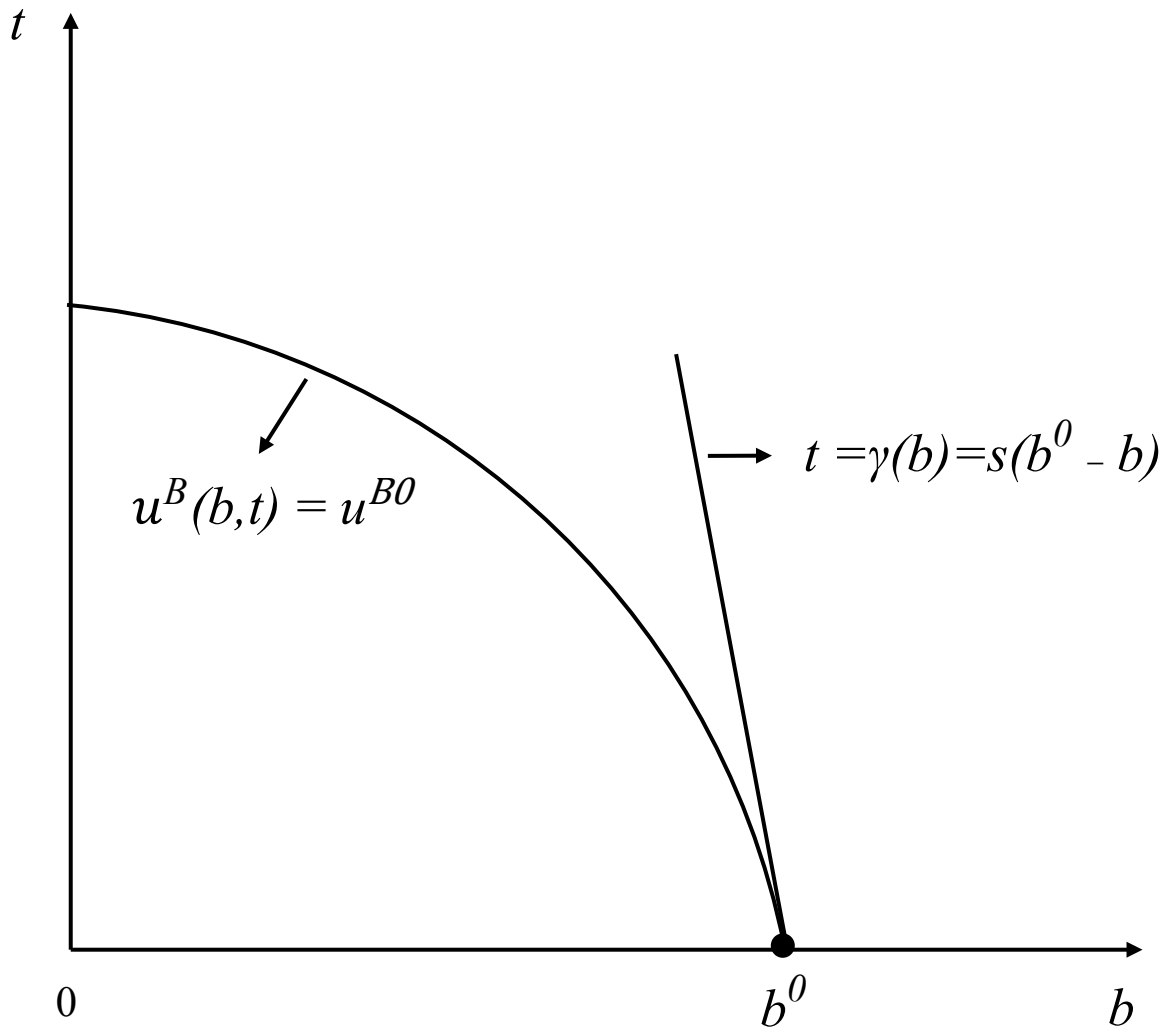

s = Absolute value of slope of B's indifference curve at $(b=b^0, t=0)$

Figure 1: Tariff threat equilibrium with reputation cost